SMRs & Green Energy: Nuclear Maritime Application

Hyun-BuKi Jeon, Director of NPR Offshore
September 1st, 2026

Agenda

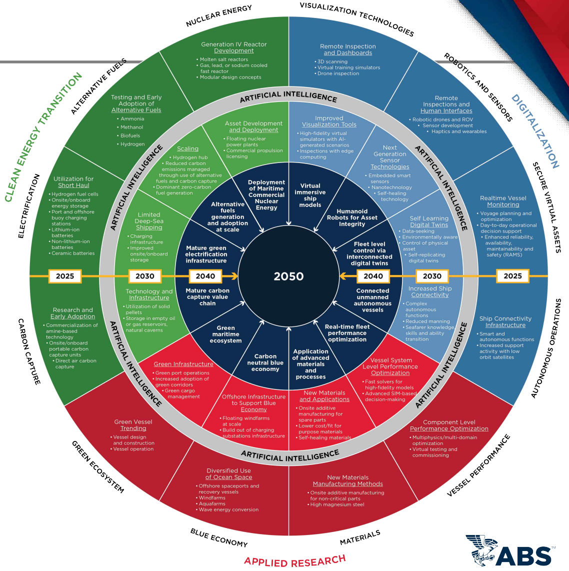
Innovation Outlook

Sustaining innovation for

Net-Zero Carbon Environment

enabled by a

Digital Ecosystem



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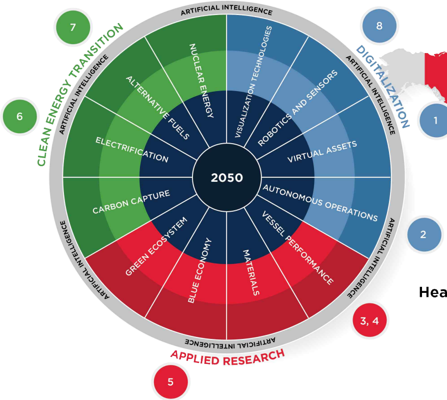
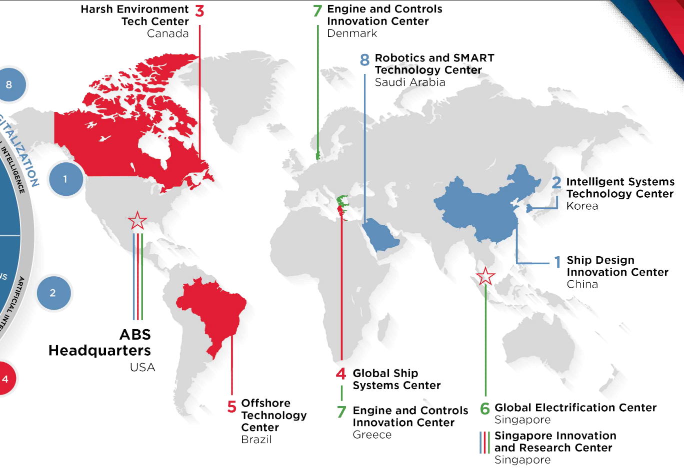
ABS Technology and Innovation Centers





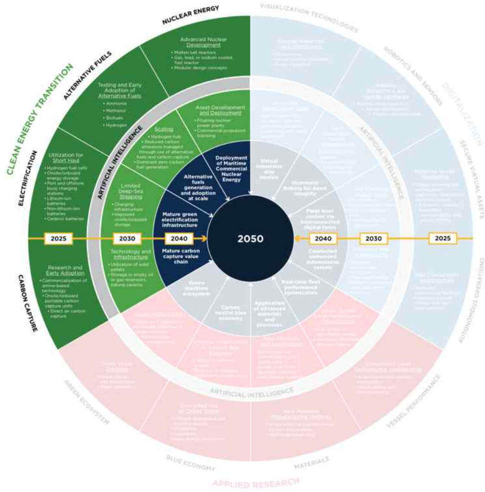
Clean Energy Transition

TECHNOLOGY ENABLERS

- Nuclear Energy
- Alternative Fuels
- Electrification Developments
- Carbon Capture Technology

DRIVERS

- Regulatory Targets
- Societal Pressures
- Finance Requirements
- Corporate Governance



Innovation Outlook

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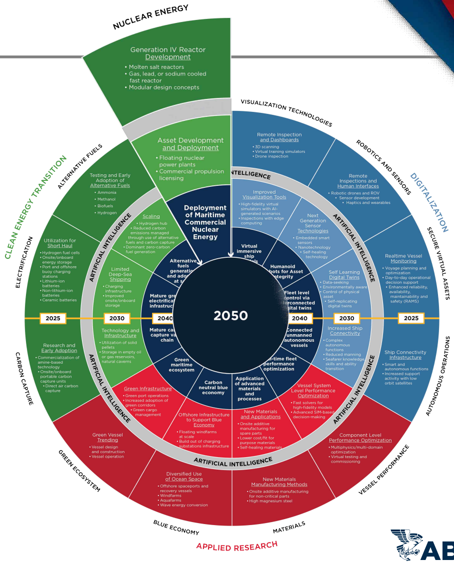
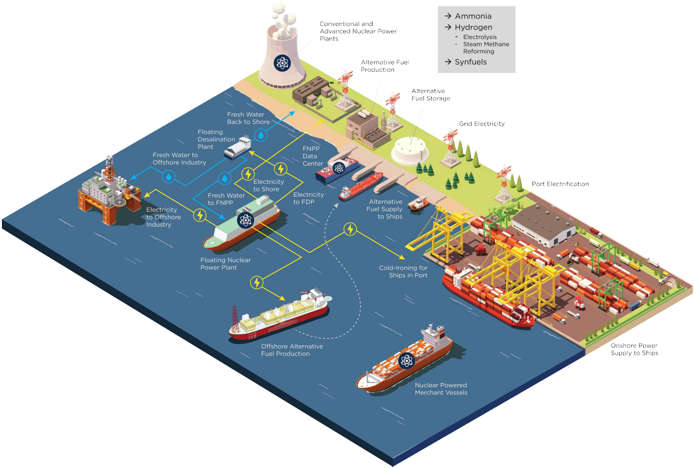


Nuclear Maritime Market



Ambition and Stopgaps for Nuclear Maritime

2050

Ambition of Energy Transition

IMO has set a goal of net zero, achievable through alternative fuels and nuclear options.

0

Adoption Gap

Today, nuclear power accounts for zero ships in the international merchant fleet.

1

Regulatory Determinant

Only one internationally recognized standard exists for nuclear merchant vessels, and it is outdated.

Requirements, Standards, Regulations

Statutory Requirements

- IMO International Code for the Safety of Life at Sea (SOLAS) CH. VIII
- IMO Resolution A.491(XII) Code of Safety for Nuclear Merchant Ships, 1981
- Other IMO Codes for Nuclear Cargo

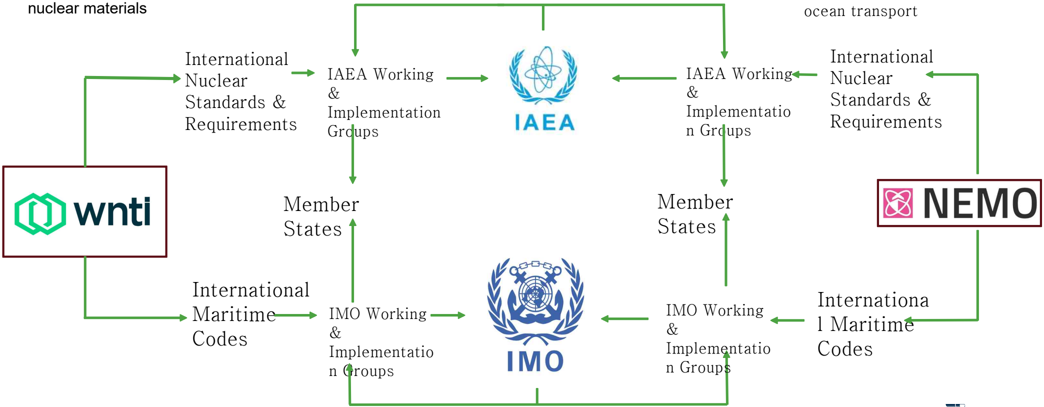
National/Industry Standards

- ABS Requirements for Nuclear Power Systems
- Nuclear Regulator (i.e., NRC) Requirements
- IAEA Standards

IMO and IAEA

Transport
Safe and secure transport of nuclear materials

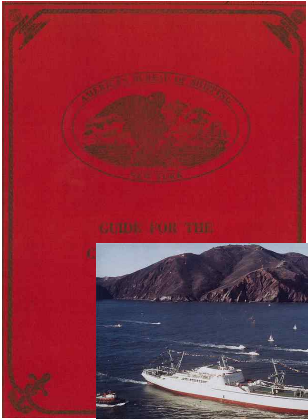
Mobility
Nuclear's role in decarbonizing ocean transport



History of ABS Nuclear Rules

ABS *Guide for the Classification of Nuclear Ships,* 1962

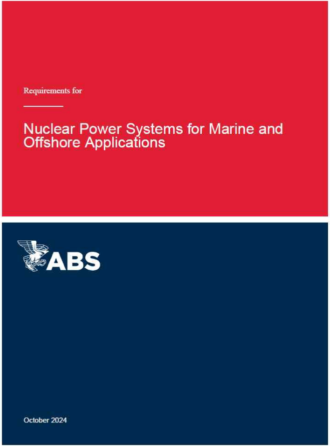
- Hull based on ABS *Steel Vessel Rules* Sections addressing:
  - Containment
  - Pressure Vessels
  - Control and Instrumentation
- Requires:
  - Nuclear Ship Safety Assessment
  - Nuclear Ship Installation Assessment
  - Consideration of credible accident scenarios
- Classed nuclear ship *Savannah* in 1961



NS SAVANNAH enroute to the World's Fair in Seattle, 1962
Credit: US Government - NARA

ABS Nuclear Requirements

- Non-nuclear propulsion maritime applications (i.e., FNPPs)
  - Ships, Barge, Floating Offshore Installation, Mobile Offshore Unit, etc.
- Mandatory **Power Service (Nuclear)** Notation
- Not applicable for review /approval of Nuclear Reactor and related systems
- Suitable for any types of nuclear technology
- Designated interface between stakeholders according to marine or nuclear service


Requirements for

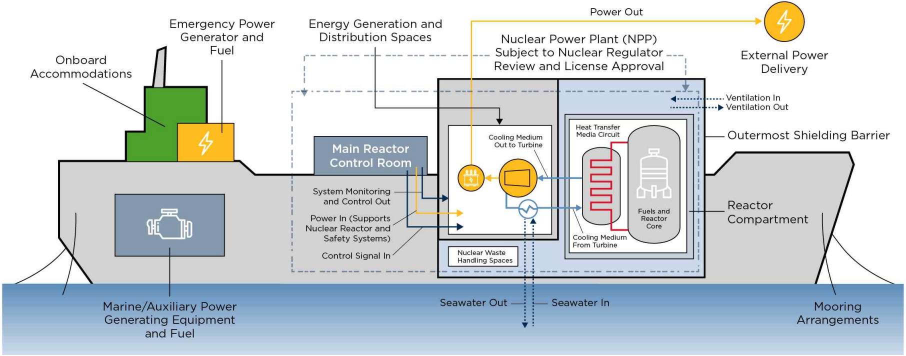
Nuclear Power Systems for Marine and Offshore Applications

ABS

October 2024

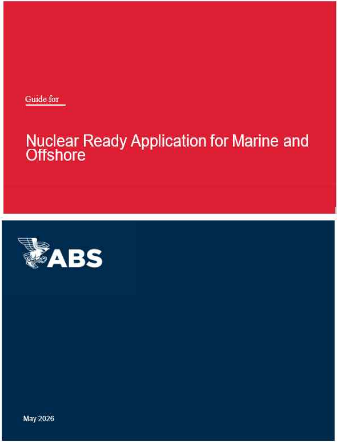
Interface Document

Establish a Technical Agreement Among Stakeholders



ABS Guide for Nuclear Ready

- Nuclear Ready maritime applications Optional Notation: **Nuclear Ready**
  - **Level 2D – Design Review**
  - **Level 3 - Installation**
- Not applicable for review /approval of Nuclear Reactor and related systems
- Nuclear technology agnostic
- Intended for vessels using a non-nuclear power source but designed for future integration of matured nuclear technology


Guide for

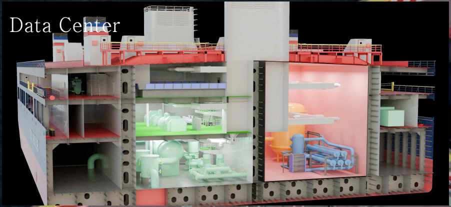
Nuclear Ready Application for Marine and Offshore

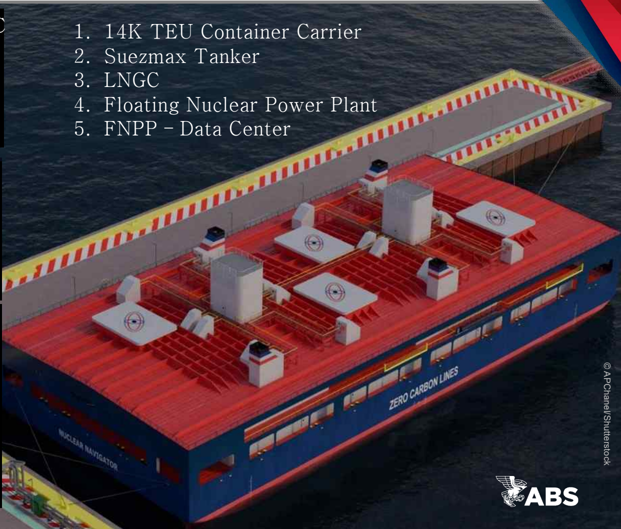
ABS

May 2026

JW0

ABS Studies on Nuclear Assets





1. 14K TEU Container Carrier
2. Suezmax Tanker
3. LNGC
4. Floating Nuclear Power Plant
5. FNPP – Data Center

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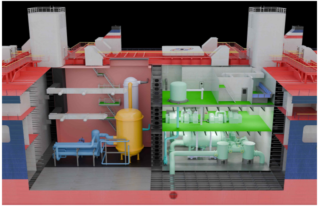
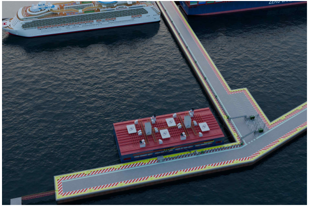
70 MWe Floating Nuclear Power Plant





POWER TO SHIPS IN PORT

- Onshore power supply to cold-iron vessels in port
- Up to 6 cruise ships at once

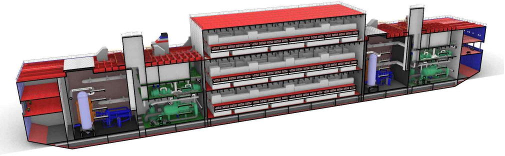
MODULAR POWER UNITS

- x 4 high temperature gas-cooled reactor (HTGR) units
- 17.5 MWe each
- 5-year fuel life

FLEXIBLE DEPLOYMENT

- 112 m length
- 50 m breadth
- Draft 4.5 m
- Installed at port, towed to location for installation

Floating Nuclear Data Center Concept



POWER TO DATA CENTERS

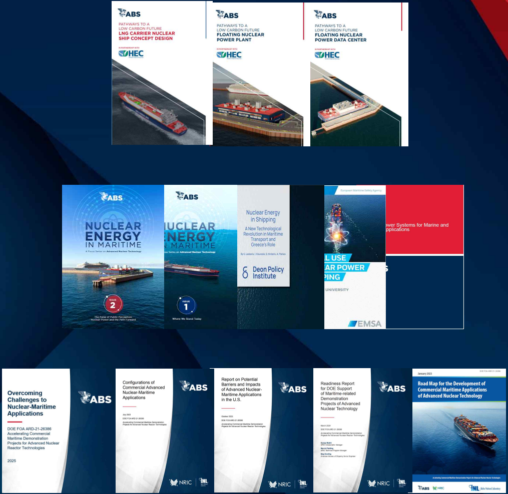
- Power direct to server racks at port
- 160 server racks @ 13.6KWe each
- 172 m length
- 50 m breadth
- Draft 6.2 m

Small Modular Reactors

- 4 high temperature gas-cooled reactor (HTGR) units
- 70 MWe total
- 5-year fuel life

ABS Publications

1. *Accelerating Commercial Marine and Offshore Demonstration Projects for Advanced Nuclear Reactor Technologies*
2. *Nuclear Energy in Maritime Focus Series*
3. *Potential Use of Nuclear Power for Shipping*
4. *ABS Requirements for Power Systems for Marine and Offshore Applications*
5. *Nuclear Energy in Shipping*
6. *Herbert concept studies*



ABS Activities and Industry Engagement

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Projects & Research

- **Herbert Eng. Concept Studies**
- **U.S. Department of Energy** *Accelerating Commercial Nuclear-Maritime Projects*
- **EMSA** *Nuclear Study*
- **Seaborg CMSR NTQ & SHI AIP**
- **HHI 15K SC-CO2 Containership AIP**
- **HHI SC-CO2 FNPP AIP**
- **KSOE Nuclear Power Barge AIP**
- **SHI 174K and 348K SMR LNGC AIPs**

ABS Supporting Services

- Classification Requirements
- Technical and Regulatory **Advisory Services**
- Technical Publications
  - **Nuclear Focus Series**
- New Technology Qualifications
- Concept & Feasibility Studies

Global Partnerships

- **EMSA**, **IACS**, **IAEA**
- World Nuclear Transport Institute (**WNTI**), Maritime Nuclear Applications Group (**MNAG**), Nuclear Energy Maritime Organization (**NEMO**)
- Universities: **Texas A&M, MIT, U. Michigan, U. Texas Dallas**

Nuclear Reactor Evolution

Generation I	Generation II	Generation III/III+

Shippingport

Diablo Canyon

AP1000

Vogtle 3 and 4

Future

First Reactor Prototypes → Commercial Energy Production → Advanced and Evolutionary Reactors → New Innovative Designs

Calder Hall (GCR/MAGNOX)
Douglas Point (PHWR/CANDU)
Desden-1 (BWR)
Fermi-1 (FBR/SFR)
Peach Bottom (HTGR)
Shippingport (PWR)
Obninsk (LWGR)

Bruce (PHRWR/CANDU)
Calvert Cliffs (PWR)
Flamanville 1-2 (PWR)
Grand Gulf (BWR)
Kalinin (PWR/VVER)
Kursk-1 (LWGR/RBMK)
Palo Verde (PWR)

ABWR (GE-Hitachi; Toshiba BWR)
ACR 1000 (AECL CANDU PHWR)
AP1000 (Westinghouse-Toshiba PWR)
APR-1400 (KHNP PWR)
APWR (Mitsubishi PWR)
Atmea-1 (Areva NP-Mitsubishi PWR)
Candu 6 (AECL PHWR)
EPR (AREVA NP PWR)

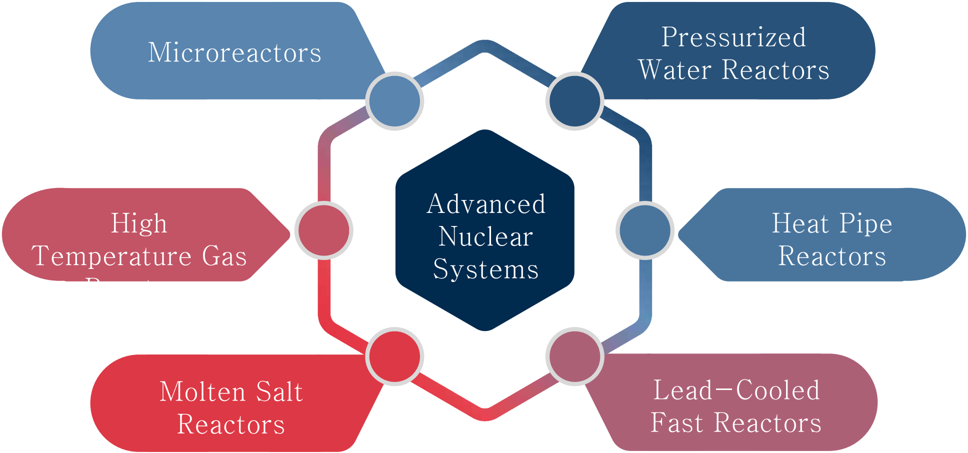
ESBWR (GE/Hitachi BWR)
Small Modular Reactors
- B&W mPower PWR
- CNEA CAREM PWR
- India DAE AHWR
- KAERI SMART PWR
- NuScale PWR
- OKBM KLT-405 PWR

VVER-1200 (Gidopress PWR)

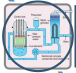
GFR	Gas-Cooled Fast Reactor
LFR	Lead-Cooled Fast Reactor
MSR	Molten Salt Reactor
SFR	Sodium-Cooled Fast Reactor
SCWR	Supercritical Water Reactor
VHTR	Very High-Temperature Reactor

1950 | 1960 | 1970 | 1980 | 1990 | 2000 | 2010 | 2020 | 2030 | 2040

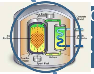
Advanced Nuclear Systems



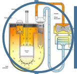
Reactor Types and Maturity



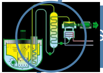
Water Cooled Reactors
- Pressurized Water Reactors
- Boiled Water Reactors



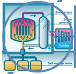
Gas Cooled Reactors
- Prismatic Gas Cooled Reactors
- Pebble-bed Gas Cooled Reactor



Liquid Metal Reactors
- Lead or LBE Cooled Reactors



Sodium Fast Reactors



Molten Salt Reactors

Reactor Type	Maturity Level	Operational Experience	Current Development Focus
Water-Cooled Reactors (PWR, BWR, PHWR)	Very High – Commercialized globally; dominant	Extensive – ~18,000 reactor-years; hundreds of units	Gen III+/SMR; optimized fuel cycles;
Gas-Cooled Reactors (AGR, HTGR)	Moderate – Some commercial use; HTGRs demonstration	Moderate – ~2,000 reactor-years; UK + China + Japan	HTGR-based SMRs (e.g., Xe-100)
Liquid Metal-Cooled Reactors (SFR, LFR)	Low to Moderate – Limited commercial use; mostly prototypical	Low – ~500 reactor-years; sodium/LBE challenges	Gen IV fast reactors, closed fuel cycles; R&D on LFRs
Molten Salt Reactors (MSR)	Very Low – Experimental; no commercial systems	Very Low – <5 reactor-years; MSRE + testbeds	Gen IV development focused on safety/fuel; early-stage concepts
All Other Reactor Types (most SMRs and Microreactors)	Conceptual	None	Component demonstration stage

Global SMR Development

Credit: NEA SMR Dashboard

MK3

Possible Collaborations

Training

Testing, validation and regulatory practice development

Joint Projects

Process development for a combined construction and installation nuclear shipyard

Business Models

Exploration of emerging business models in the nuclear–maritime industry

Advisory Services

Techno–economic review of multiple nuclear–maritime topics of inquiry

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"We have not done this recently, but we have done this before."

— NRIC

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